BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: PR–XII (A-I-B) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR–XII (A-I-B) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. An alternating voltage is given by 20Sin(157t), the frequency of AC is:

(A) 25 Hz
(B) 50 Hz
(C) 75 Hz
(D) 100 Hz

Q2. In Hysteresis, magnetic flux density (B) that remains in material when magnetizing force is zero is termed as;

(A) Retentively
(B) Reluctance
(C) Coercive force
(D) Residual flux

Q3. In reversed biased condition of a diode, sudden increase in current is known as ____effect.

(A) Knee
(B) Compton
(C) Zener
(D) Photoelectric

Q4. The wavelength of electrons is ____times shorter than visible light.

(A) 10^3
(B) 10^5
(C) 10^7
(D) 10^9

Q5. According to uncertainty principle, if uncertainty of position of electron becomes 3 times, then uncertainty in momentum will be:

(A) 3 times
(B) 1/3 times
(C) 9 times
(D) 1/9 times

Q6. The first excitation potential of hydrogen atom is:

(A) 219.5 volt
(B) 12.1 volt
(C) 10.2 volt
(D) 0 volt

Q7. Helium-Neon Laser Beam emitted from a discharge tube has ____colour.

(A) Blue
(B) Green
(C) Red
(D) White

Q8. In nucleus, beta (β) emission does not change the number of:

(A) Nucleon
(B) Proton
(C) Neutron
(D) None of these

Q9. All free particles have colour charge of:

(A) 0
(B) 2/3
(C) 1/3
(D) 1

Q10. The time constant (RC) has the same unit as that of:

(A) Resistance
(B) Charge
(C) Capacitance
(D) Time

Q11. Object 'A' and 'B' have charges 2μC and 6μC respectively. Which statement is true about the electric forces on the objects?

(A) F_AB=F_BA
(B) F_AB=-3F_BA
(C) -3F_BA=F_BA
(D) F_AB=-F_BA

Q12. If slope of an Ohmic substance form an angle of 30° in I-V graph, then resistance of the conductor will be:

(A) 0Ω
(B) √3Ω
(C) 1/√3Ω
(D) Infinite

Q13. To obtain maximum output power, resistance of source is equal to:

(A) Load resistance
(B) Infinite
(C) Zero
(D) Constant

Q14. When an resistance each of value "r" are connected in parallel, then resultant resistance is x. When these n resistances are connected in series, then total resistance is:

(A) nx
(B) Rnx
(C) x/n
(D) n^2x

Q15. Which of the given has the least resistance in the measuring instruments;

(A) Galvanometer
(B) Voltmeter
(C) Ammeter
(D) All have the same

Q16. When north pole of a magnet move towards the coil, then that end of the coil will act as:

(A) Positive charge
(B) Negative charge
(C) North pole
(D) South pole

Q17. The SI unit of mutual inductance (M) is:

(A) Farad
(B) Weber
(C) Tesla
(D) Henry

Q18. In pure inductive circuit, the phase difference between I and V is:

(A) 0
(B) π/2
(C) π
(D) 3π/2

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FOUR marks.

1) Show that Coulomb's Law fits into Newton's 3rd Law of Motion?

2) How are units of volts and electron volts related? How do they differ?

3) A heavy duty battery of truck maintains a current of 3A for 24 hours. How much charge flows from the battery during this time?

4) Calculate the resistance of a wire 10m long that has a diameter of 2mm and resistivity of 2.63×10^{-8} Ωm.

5) How does a current carrying coil behave like a bar magnet?

6) How electromagnetic induction is used in cook tops in electric ranges?

7) The value of A.C voltage across 0.5 μF capacitor is $16\text{Sin}(2\text{x}10^3)$V. Find capacitive reactance?

8) Coercive force of steel is greater than iron. Why?

9) Why the base current is weak as compare to collector current?

10) Write a note on pair annihilation.

11) Write down few applications of x-rays.

12) Describe the interaction of radiations with matter?

13) What is nuclear fission chain reaction? Explain.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Attempt any THREE questions. All questions carries equal marks.

Q. Attempt any THREE questions. All questions carries equal marks.

Q3.

(a) Define and explain electric potential energy and electric potential due to a point charge.

(b) Two positive charges of $15x10^{-10}C$ and $13x10^{-10}C$ are placed 12cm apart. Find the work done bringing the charges 4cm closer.

Q4.

(a) Explain construction and working of AC generator.

(b) Current in a circuit falls from 5.0A to 0A in 1s. If an average emf of 200V induced, calculate self inductance of the coil?

Q5.

(a) What is Half life of a radio-active nucleus? Derive mathematical relation for it.

(b) The temperature of the human skin is 35°C. What is the wavelength at which the peak occurs in the radiation emitted from the skin?

Q6. Give a brief explanation of any two topics.

(a) Determine the electric field intensity due to infinite sheet of charge

(b) Explain RLC Series AC circuit in detail.

(c) Describe Compton's Effect